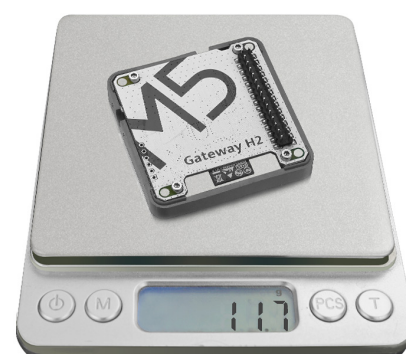


Module Gateway H2

SKU: M141



Description

The **Module Gateway H2** is a stackable gateway development module based on the ESP32-H2-MINI-1 module, specifically designed for the M5 series hosts. It features a low-power, **RISC-V**-based 32-bit single-core MCU and supports **Zigbee**, **Thread**, and the Thread-based Matter interconnect protocol. The module integrates **IEEE 802.15.4** wireless communication capabilities, enabling it to work with M5 hosts to build Matter over Thread end devices, facilitating interoperability across multiple ecosystems.

Additionally, the Module Gateway H2 includes a hardware encryption engine, meeting the data security requirements of high-demand IoT applications. The module is equipped with a DIP switch, allowing users to flexibly switch communication interface modes, and it also provides a standard program download interface for easy firmware burning and debugging. It supports multi-module stacking expansion, making it suitable for applications such as smart homes, environmental monitoring, sensor networks, and low-power wireless communication nodes, helping developers quickly prototype and develop products.

| Features

- Support for wireless communication protocols such as Zigbee, Thread, and Matter
- Integrated IEEE 802.15.4 technology
- DIP switch configuration
- Hardware encryption support
- Stackable expansion
- Low power consumption
- Development platforms:
 - Arduino IDE
 - ESP-IDF

| Includes

- 1 x Module Gateway H2

| Applications

- Smart home
- Industrial automation
- Healthcare
- Consumer electronics
- Smart agriculture

| Specifications

Specification	Parameters
SoC	Built-in ESP32-H2 chip, RISC-V 32-bit single-core processor, up to 96 MHz @IEEE 802.15.4
Flash	2MB
Antenna	Onboard PCB antenna
Supported Protocols	IEEE 802.15.4 (including Zigbee 3.0, Thread 1.3, Matter)
Power Consumption	Standby current: DC3.3V/8.55mA Operating current: Thread networking: DC3.3V/18.35mA
DIP Switch	8-bit DIP switch for interface switching (UART/SPI) and wireless function control
Operating Temperature	0 ~ 40℃
Product Size	54.0 x 54.0 x 13.1mm
Product Weight	11.7g
Package Size	132.0 x 95.0 x 16.0mm
Gross Weight	25.3g

Learn

Downloading Programs

Use the [ESP32 Downloader](#) to download programs to the ESP32-H2.



Certifications

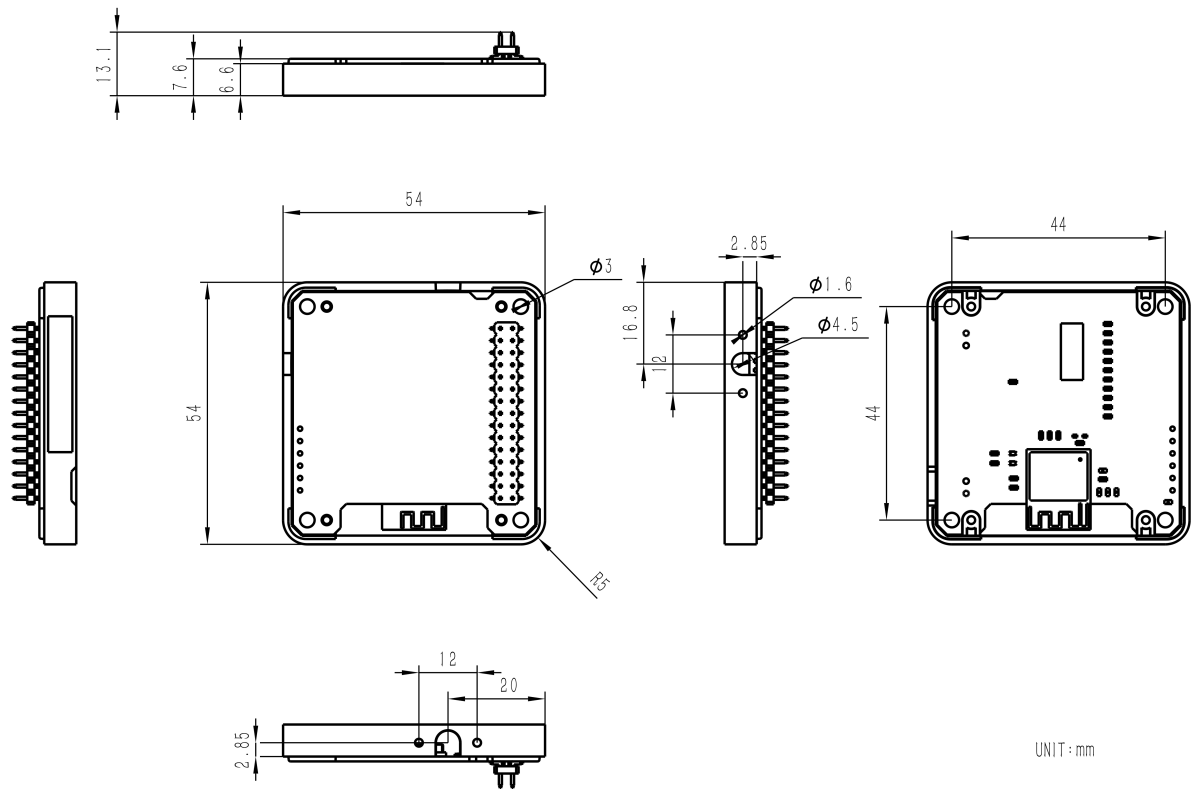
- RCM/TELEC/CE/FCC/IC/BQB certifications

Schematics

- [Module Gateway H2 Schematic PDF](#)

PIN	LEFT	RIGHT	PIN
GND	1	2	RXD
GND	3	4	
GND	5	6	
SPI_MOSI	7	8	BT_PRIORITY
SPI_MISO	9	10	
SPI_CLK	11	12	3V3
	13	14	
G9	15	16	
	17	18	
	19	20	
BT_ACTIVE	21	22	H2-EN
SPI_CS	23	24	WL_ACTIVE
HPWR	25	26	
HPWR	27	28	5V
HPWR	29	30	BAT

Model Size



Datasheets

- [ESP32-H2-MINI-1 Module Datasheet](#)

Softwares

Arduino

- [Module Gateway H2 Arduino Tutorial](#)

UiFlow2

- [Module Gateway H2 UiFlow2 Docs](#)

ESP-IDF

- [ESP Thread Border Router Guide](#)
- [ESP Zigbee Gateway Guide](#)
- [ESP Zigbee NCP Guide](#)
- [ESP32-H2 ESP-IDF Docs](#)
- [ESP Zigbee SDK](#)
- [ESP Thread Border Router SDK](#)

| Video

- Module Gateway H2 Product Introduction and Case Demonstration

[M141_Module_Gateway_H2_video.mp4](#)